

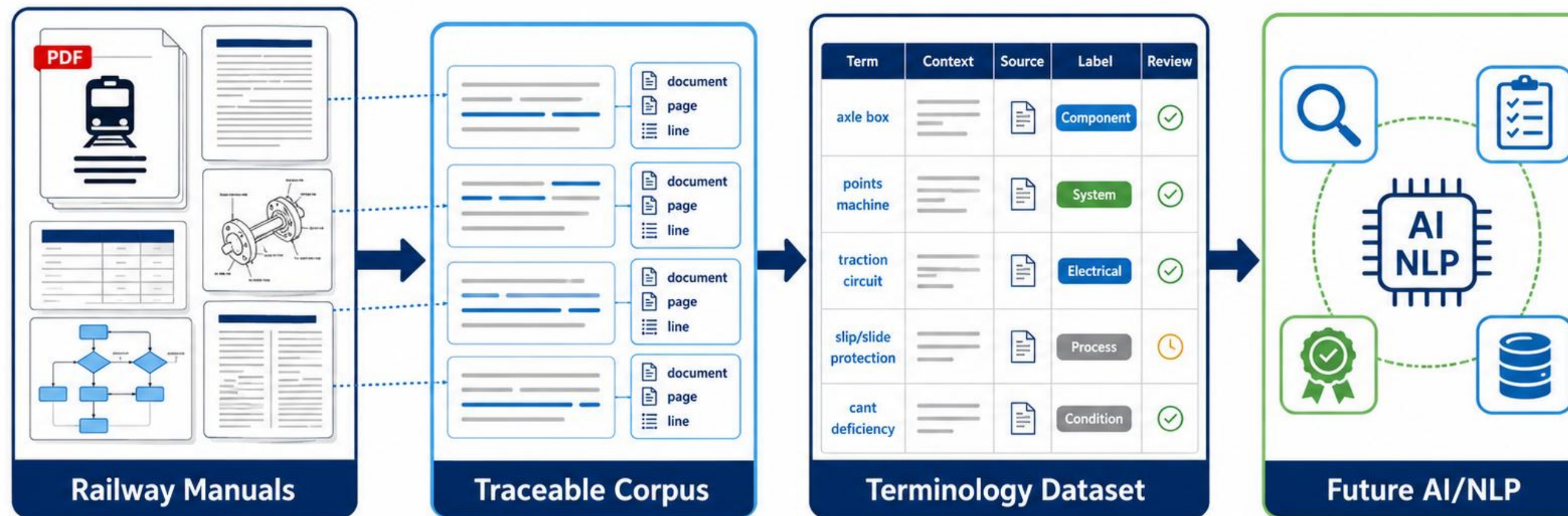


MONTHLY RESEARCH REPORT

	Month	Month 1 / From 01/05/2026 – 01/06/2026
	Member	AKIRA, SHUICHI
	Research Theme	Building a Railway Terminology Training Dataset
	Mentor	Dr. BUI QUOC BAO

ONE LOVE. ONE FUTURE.

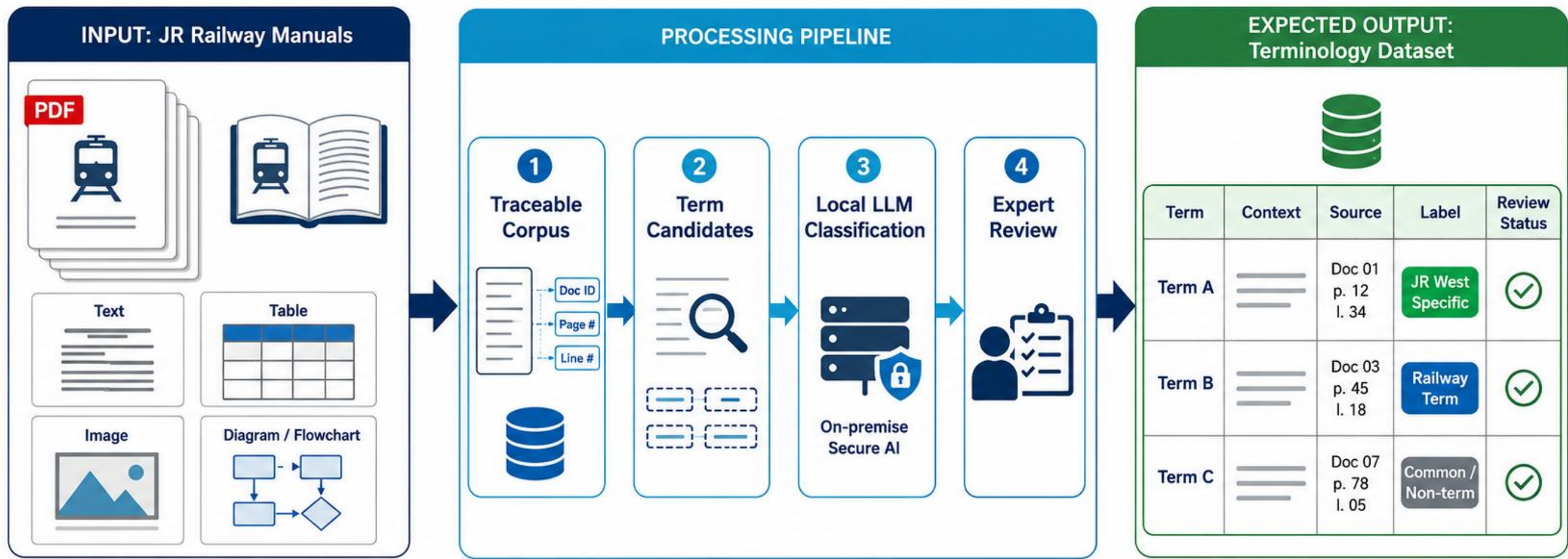
Project Vision: From Railway Manuals to Terminology Dataset



Traceable terminology data for review, standardization, and future AI/NLP systems

Transform JR railway manuals into a traceable terminology dataset for review, standardization, and future AI/NLP systems.

Project Overview



 A reliable terminology foundation for review, standardization, and future AI/NLP applications. 

The project converts JR railway manuals into a structured, traceable terminology dataset.

System Architecture

A phased pipeline that converts railway manuals into reviewed terminology data.

Phase 1

Document Preprocessing
Input: PDF Manuals
Output: Traceable Corpus

- Docling conversion
- OCR integration
- Page-line traceability



Phase 2

Candidate Extraction
Input: Traceable Corpus
Output: Term Candidates

- GiNZA extraction
- BERT/RoBERTa scoring
- Noise filtering



Phase 3

Local LLM Classification
Input: Context
Output: Term Labels

- JR West specific
- Railway term
- Common / Non-term



Phase 4

Human Review & Dataset Delivery
Input: Classified Terms
Output: Final Dataset

- Expert validation
- Criteria refinement
- Review status



Traceability is preserved across all phases: document → page → line → term.

The focus for May 2026 is centered on expanding high-precision data ingestion and refining terminology engineering. We aim to establish a seamless, phased pipeline that converts legacy railway manuals into reviewed training data.



PDF-to-Corpus Pipeline

Convert PDF manuals into clean, traceable corpus.

- PDF conversion
- OCR integration
- Page-line traceability



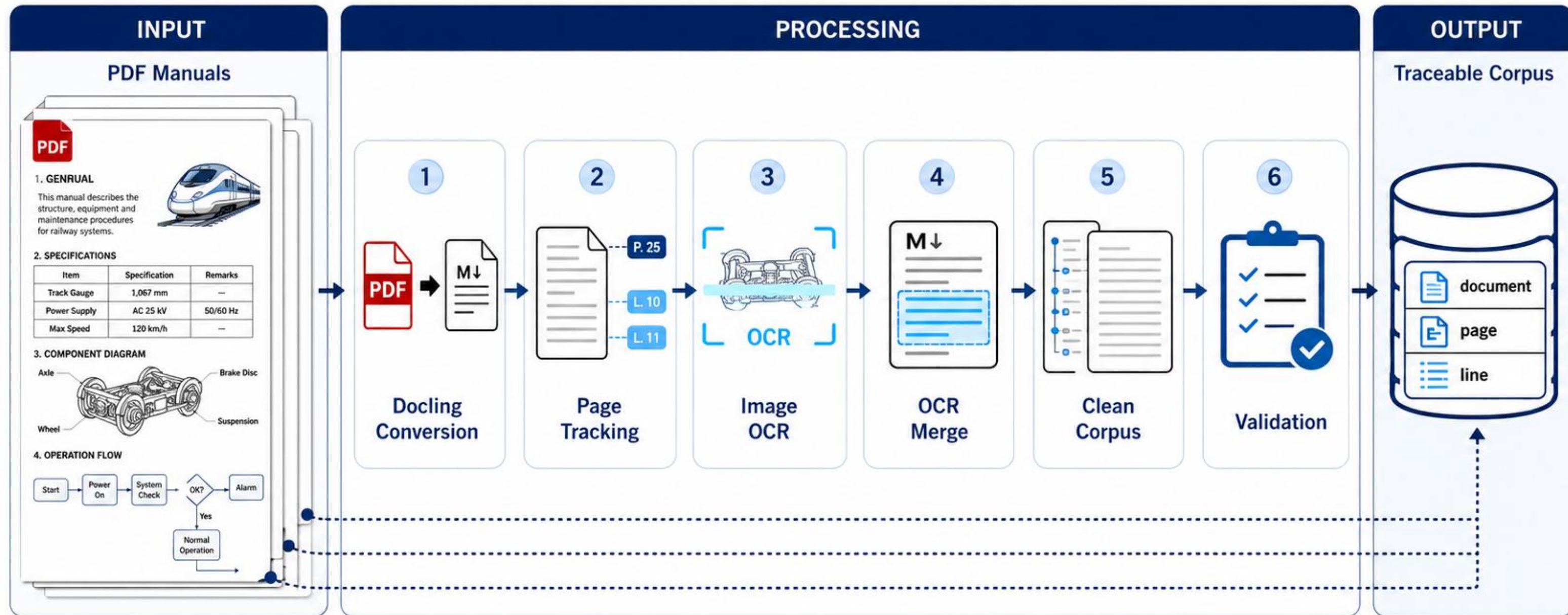
Candidate Term Extraction

Generate candidate railway terms from the corpus.

- GiNZA extraction
- Weak labeling
- BERT/RoBERTa scoring

Phase 1: PDF to Traceable Corpus

Building a clean and source-traceable text foundation before term extraction.



Output: Clean text with OCR content and document-page-line traceability.

Phase 1 Example: Source-Traceable Text Extraction

Extracted text can be verified against the original PDF page.

Text extracted from PDF

ドア「閉」とするも運転士知らせ灯(または戸ジメ表示灯)が点灯しない
または、ドア「開」とするもドアが開扉しない
運転士知らせ灯が点灯しない場合、以下の処置を行い運転士知らせ灯が点灯すれば、運転再開
運転台選択SWの確認
再度、ドアの開閉動作

Matched

☐ ドア「閉」とするも運転士知らせ灯（または戸ジメ表示灯）が点灯しない
または、ドア「開」とするもドアが開扉しない

☐ 運転士知らせ灯が点灯しない場合、以下の処置
を行い運転士知らせ灯が点灯すれば、運転再開
・運転台選択SWの確認
・再度、ドアの開閉動作

OCR text from image region

連絡
以下の3点を連絡
1 運転士知らせ灯の点灯状態
2 車側灯の点灯状態
3 ドアの開閉状況

Matched

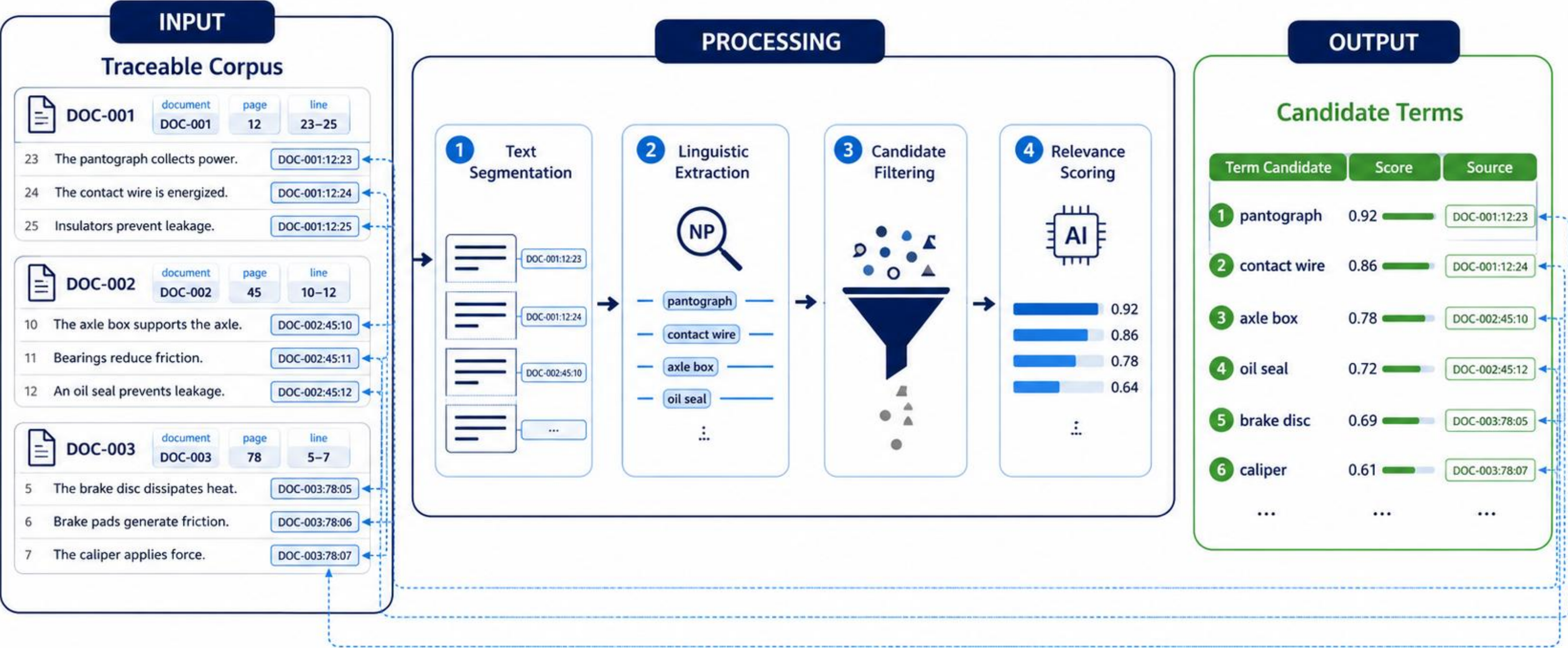
連絡

☐ 以下の3点を連絡
① 運転士知らせ灯の点灯状態
② 車側灯の点灯状態
③ ドアの開閉状況

Matched source region

Phase 2: Candidate Term Extraction

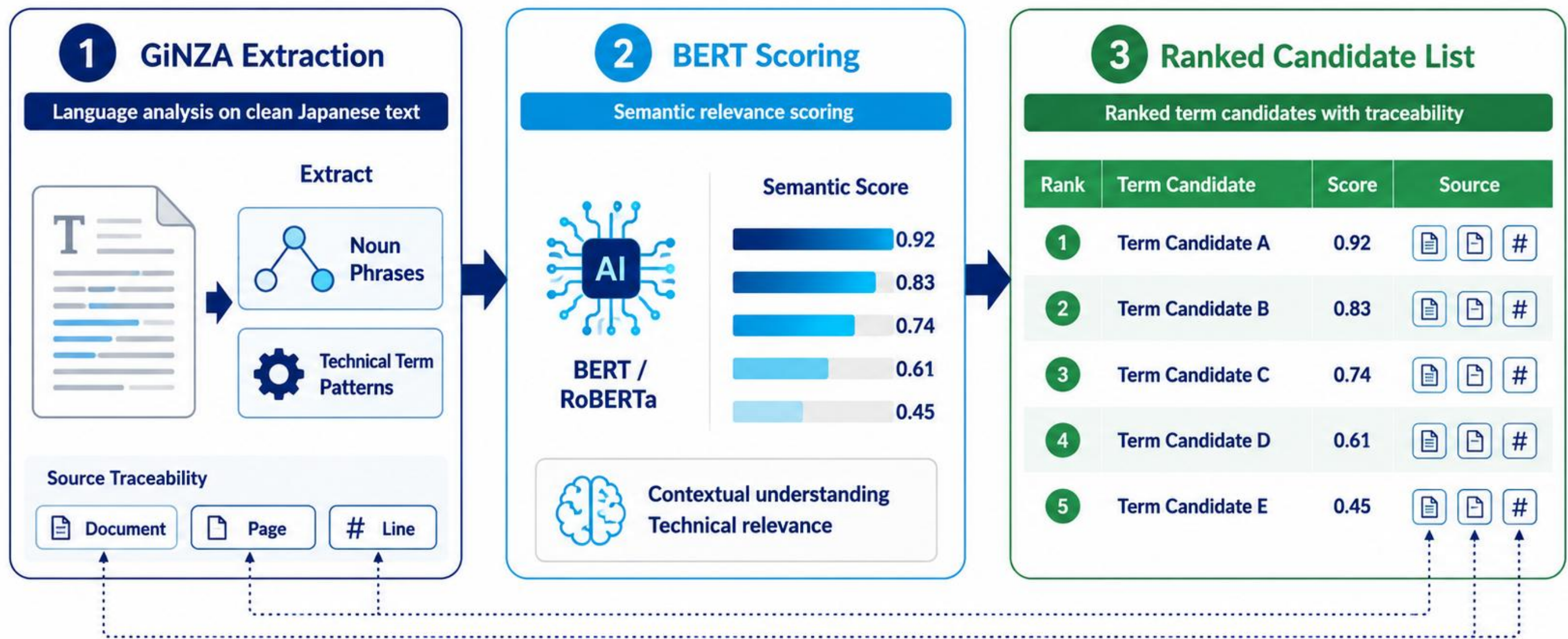
Turning the traceable corpus into ranked terminology candidates.



Output: Ranked candidate terms with source traceability.

Candidate Extraction Baseline

Combining linguistic extraction with contextual scoring.



Model	Precision	Recall	F1	Accuracy
cl-tohoku/bert-base-japanese-v3	0.8658	0.9474	0.9281	0.9355
rinna/japanese-roberta-base	0.8814	0.9316	0.9170	0.9232

Term Candidate Output

Each candidate keeps both relevance score and source traceability.

cleaned_candidate	term_group	review_priority	avg_score	source_files	source_refs	example_sentence
電空開閉器	new_potential_term	1	0,999283087	ber_ready(2).txt	new_doc_2 line:3638, new_doc_2 lin	電空開閉器「連結」位置
主幹制御器	new_potential_term	1	0,99915638	ber_ready.txt, _1-05_()(2025.09.18)_c	new_doc_main line:3019, new_doc_n	主幹制御器
連結操作スイッチ	new_potential_term	1	0,999111333	ber_ready(2).txt	new_doc_2 line:3606, new_doc_2 lin	連結操作スイッチ「連結」とし、
電源誘導	new_potential_term	1	0,99906022	ber_ready.txt, ber_ready(2).txt	new_doc_main line:2539, new_doc_2	電源誘導(発生時刻:)
切換スイッチ	new_potential_term	1	0,998826002	ber_ready(3).txt, ber_ready.txt, ber_r	new_doc_3 line:318, new_doc_3 line	1所定停止位置に停止2車掌スイッチ「閉」確認(両側)3切換スイ
隣接線抑止	new_potential_term	1	0,998974276	_1-05_()(2025.09.18)_canonical(1)_v	conductor_manual_1_05 line:2004, r	隣接線抑止が必要な
遮断桿	new_potential_term	1	0,998893142	_1-05_()(2025.09.18)_canonical(1)_v	conductor_manual_1_05 line:962, co	遮断桿の折損等発生時の徐行による列車影響を最小限に抑える
所定停止位置	new_potential_term	1	0,998404599	ber_ready.txt, _1-05_()(2025.09.18)_c	railway_ocr_new line:1983, railway_o	通過列車を含めて所定停止位置に停止
停止位置	new_potential_term	1	0,998391541	ber_ready.txt, _1-05_()(2025.09.18)_c	railway_ocr_new line:2577, conducto	指令員又は駅長に信号機を目標物として停止位置を報告
EB装置	new_potential_term	1	0,998297006	ber_ready.txt	railway_ocr_new line:3160, new_doc	EB装置の場合は速度検出器
隣接線支障	new_potential_term	1	0,998576854	_1-05_()(2025.09.18)_canonical(1)_v	conductor_manual_1_05 line:671, rai	隣接線支障の場合は列車防護
非常停止	new_potential_term	1	0,998614625	ber_ready(3).txt, _1-05_()(2025.09.18)	new_doc_3 line:397, new_doc_3 line	(5)非常停止のときは直ちに自動ブレーキの「非常位置」をとる
三相表示灯	new_potential_term	1	0,998675579	ber_ready.txt, ber_ready(3).txt, ber_r	new_doc_main line:303, new_doc_3	※2指令への連絡事項として、力行中・ダ行中・ブレーキ中等
会社間表示灯	new_potential_term	1	0,998912066	_1-05_()(2025.09.18)_canonical(1)_v	conductor_manual_1_05 line:385, co	会社間表示灯
退行運転	new_potential_term	1	0,998802597	_1-05_()(2025.09.18)_canonical(1)_v	conductor_manual_1_05 line:1272, c	退行運転
逆転ハンドル	new_potential_term	1	0,9977931	ber_ready.txt, ber_ready(3).txt	new_doc_main line:1896, new_doc_n	※主制御器MF化車両は逆転ハンドル「切」位置
直下地上子	new_potential_term	1	0,998101652	ber_ready.txt	railway_ocr_new line:3024, railway_o	直下地上子を越える / ブレーキ開放ボタンを押す機外に一旦停
緊急停止	new_potential_term	1	0,998234799	_1-05_()(2025.09.18)_canonical(1)_v	conductor_manual_1_05 line:1828, r	(3)photo shareによる運転取扱いについては、アーバンエリアブ
無線機電源	new_potential_term	1	0,998269115	ber_ready(3).txt	new_doc_3 line:1040, new_doc_3 lin	無線機電源を「入」とする電源切換SWの「常用」位置を確認
事故表示灯	new_potential_term	1	0,998567043	ber_ready.txt, ber_ready(2).txt, _1-05	new_doc_main line:1588, new_doc_n	運転再開後、再度事故表示灯点灯するか?
ノッチ	new_potential_term	1	0,998074391	ber_ready(3).txt, ber_ready(2).txt, be	new_doc_3 line:428, new_doc_3 line	(3)抑速ノッチを進める必要があるときは、常用ブレーキにより
室内灯	new_potential_term	1	0,997958388	ber_ready(2).txt, ber_ready.txt	new_doc_2 line:3497, new_doc_2 lin	全て室内灯点灯しているのに、三相表示灯の点灯
停止区間	new_potential_term	1	0,99806329	_1-05_()(2025.09.18)_canonical(1)_v	conductor_manual_1_05 line:1792, r	自列車の停止位置が赤区間(停止区間)
赤色灯	new_potential_term	1	0,998440356	_1-05_()(2025.09.18)_canonical(1)_v	conductor_manual_1_05 line:858, rai	異常挙動検知時は上の白色灯が点灯故障発生時は下の赤色灯が
白色灯	new_potential_term	1	0,998324782	_1-05_()(2025.09.18)_canonical(1)_v	conductor_manual_1_05 line:858, rai	異常挙動検知時は上の白色灯が点灯故障発生時は下の赤色灯が

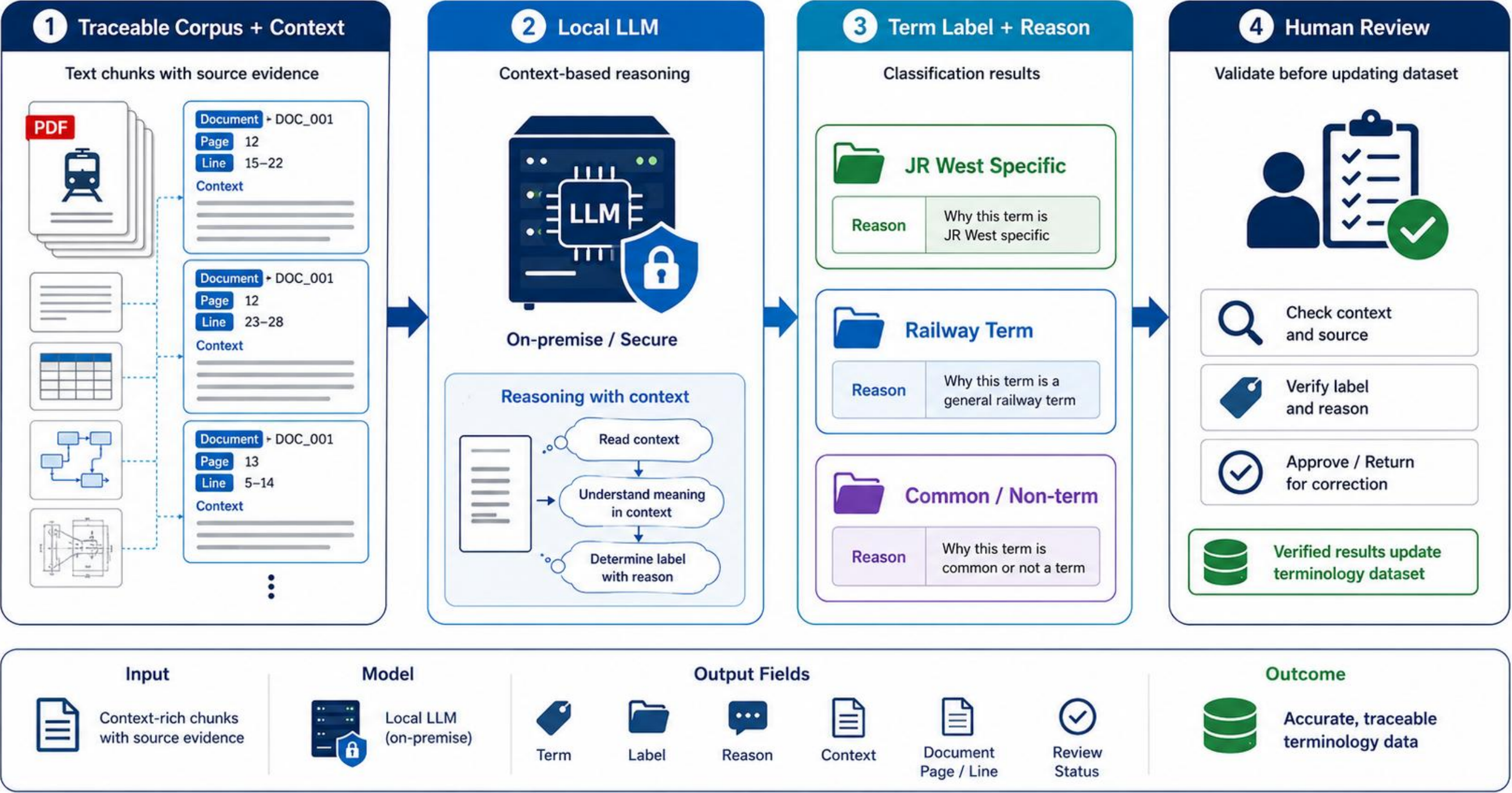
Extracted Candidates

Bert's Confidence Score for Candidates

Candidates' source traceability (file names + line + sentences including candidates)

Next Phase: Local LLM Classification

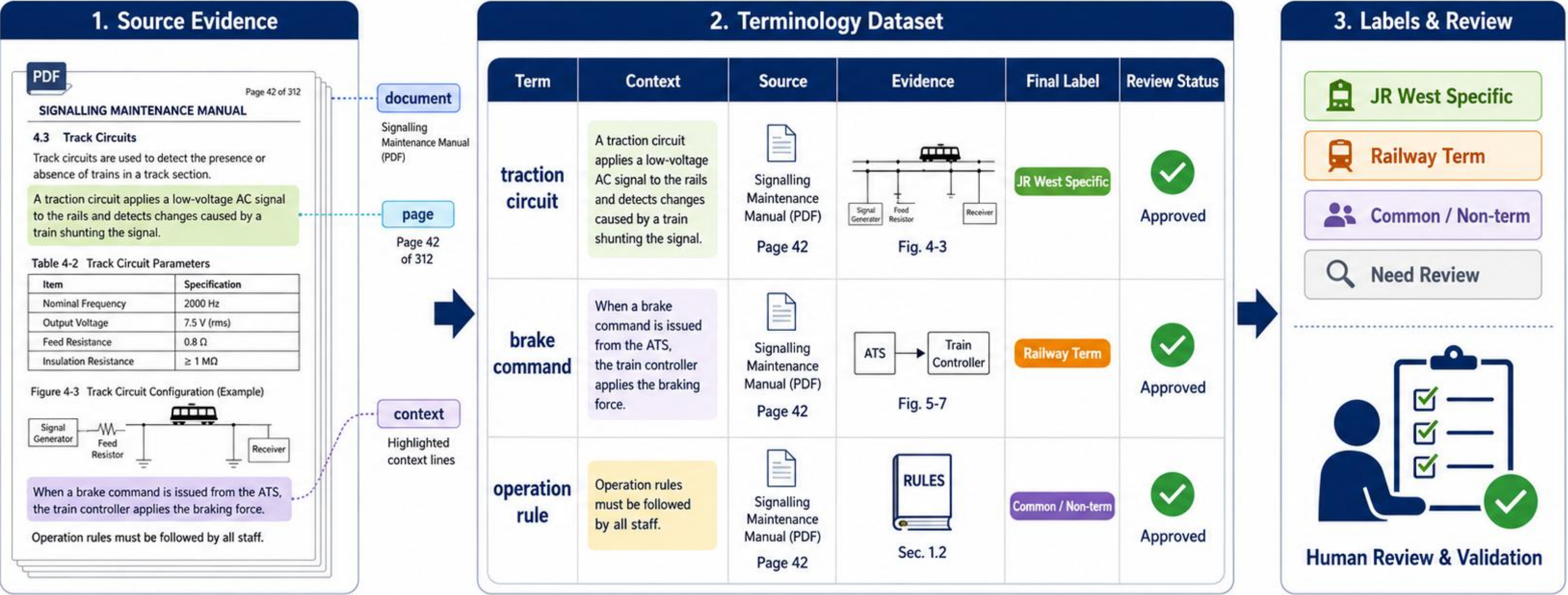
Classify terms by reading their context, not as isolated words.



Traceable Corpus + Context → Local LLM → Term Label + Reason → Human Review

Final Output: Railway Terminology Dataset

Each term is reviewable, explainable, and traceable to its original



Source traceability:



Research Roadmap

Planned work from corpus completion to dataset finalization.



The HUST logo is centered on a dark blue background. It consists of the letters 'HUST' in a bold, white, sans-serif font. Surrounding the text is a large, circular graphic made of many small red dots, arranged in a pattern that resembles a stylized flower or a sunburst.

HUST

